This certificate replaces the electronic certificate HAM2303402/1/A2, dated 24 September 2024, which is hereby cancelled

Certificate for AC Generators or Motors

Office	
Hamburg	
Client	Date
VEM Sachsenwerk GmbH **Dresden - Germany**	**06 June 2025**
	Order number on Manufacturer

	Work's order number
	K-2200120
Manufacturer	Intended for
VEM Sachsenwerk GmbH	**Besecke GmbH**
First date of inspection	Final date of inspection
25 October 2023	**25 October 2023**

This certificate is issued to the above Client to certify that the ac generator/motor, particulars of which are given below, has been inspected at the manufacturer's works. The construction, workmanship and materials are good, and the machine complies with the relevant requirements of the Rules and Regulations.
On completion the generator/motor was tested with satisfactory results.

Particulars

Type			
Auxiliary AC Generator ☒	Auxiliary AC Motor ☒	Propulsion AC Generator ☐	Propulsion AC Motor ☐
kVA (generator only)	Volts	Number of phases	kW
325	**520/690**	**3**	**600/600**
Amperes	Hertz	Power factor	Rev/min
726/564/361*	**40/73.3**	**0.94/0.91/0.92***	**1200/2200**
Type of enclosure		Class of insulation	
IP54		**180(H) used 155(F)**	
Type number	Serial number	Date of temperature test	Machine acting as
P62B 355LX4	**45615803**	**25 October 2023**	**PTI/PTO Motor/Generator**

Results Of Tests

Duration	Rev/min	Volts	Amperes
360'	**1200**	**218.2**	**725.3**
Hertz	Power factor	Field-volts	Field-amperes
40	**0.0.34**	**---**	**---**

DEGREES C (State whether resistance ("r") or thermometer ("t")

Test	Actual	Rise
Cooling Air	**21.5 ambient**	**---**
Cooling Water	**26.1**	**27.9**
Stator Winding	**76.3**	**50.2**
Rotor Winding	**---**	**---**
Slip Rings	**---**	**---**

Generator Voltage Regulation

If Regulation not inherent state serial number of A.V.R

Test	Full load	No load
Rev/Min	**---**	**1199**
Volts	**---**	**510**
Amperes	**---**	**no load**

Hot insulation resistance (megaohms)	High voltage test volts ac for 1 minutes	Overload test
14900	**2.4kV**	**1.1 x In 801A (15'); overspeed test 2640 rpm (120s)**

Identification Marks
Mark "n/a" if not applicable

Identification number (including office contraction code)

LR TEST HAM 2303402/1 10 OR 23, 45615803

Surveyor's initials	Date of inspection
OR	**25.10.2023**

*** machine act as generator.**
Type test. Shaft No.: HAM2303402/3. For details see manufacturer's test report no. 45615803 dated 25.10.2023 and Type test report dated 07.12.2023

Olaf Reger
Hamburg Office
06 June 2025
Lloyd's Register EMEA
LR425.2022

Senior Surveyor to
A subsidiary of Lloyd's Register Group Limited

Inspection certificate
Prüfbescheinigung

Inspection certificate 3.2 according to EN 10204
Abnahmeprüfzeugnis 3.2 nach EN 10204

Synchronous motor
Synchronmotor

VEM

Rated data / Bemessungsdaten

3ph Mot.	Typ P62B 355LX4		
	Nr. 45615803 / 2023		
Y	520/690 V	40.0/73.3 Hz	
	726/564 A		
	600/600 kW	n/a °C	
	cos φ 0.94/0.91	UVW	
	1200/2200 rpm	IP 54	IC 71W
permanent magnet excited		2825 kg	
Water 120 l/min max. 38 °C			
Th.cl. 180(H) used 155(F)			
LR HAM2303402/1			

General data / Allgemeine Angaben

Project title / Projektbezeichnung	My Boardwalk - PTI/PTO STB
Customer / Kunde	Besecke GmbH
Internal order no. / PSP-Nr.	K-2200120
Rotor no. / Läufer-Nr.	n/a
Winding instruction / Wickelanweisung	n/a
Exciter machine type / Erregermaschinentyp	n/a

Ex-protection data / Ex-Schutz-Angaben

Ex-certification / Ex-Zertifikat	n/a
Ex-type / Ex-Typ	Ex n/a

P_{sup} [bar]	n/a	Q_{purg} [m³/h]	n/a	t_{purg} [min]	n/a
P_{min} [mbar]	n/a	P_{max} [mbar]	n/a	$P_{ref,min}$ [mbar]	n/a
Q_{Leak} [m³/h]	n/a	$P_{ref,Leak}$ [mbar]	n/a		

Machine construction data / Angaben zur Maschinenkonstruktion

Type of construction / Bauform	IM 1001	Brushes per ring / Bürsten je Ring	n/a	Type of bearing / Lagerart	antifriction bearing	Type of grease / Fettsorte	ASONIC GHY 72
Cooling design / Kühlart	IC 71W	Brush grade / Bürstenmarke	n/a	Bearing DE / Lager DS	6324MC3	Type of oil / Ölsorte	n/a
Air gap / Luftspalt [mm]	n/a	Brush dimensions / Bürstenmaße [mm]	n/a	Bearing NDE / Lager NS	6317MC3	Oil quantity DE / Ölmenge DS [l/min] / [l]	n/a
Air-gap exciter machine / Luftspalt Erregermaschine [mm]	n/a	Slip ring serial number / Schleifringnummer	n/a	x-dimension / x-Maß [mm]	n/a	Oil quantity NDE / Ölmenge NS [l/min] / [l]	n/a

ES 02, ES 03, ES 44, SE 01 - DC & insulation resistance, Withstand voltage / DC- & Isolationswiderstände, Stehspannung

Winding / Wicklung	Number of poles / Quantity / Polzahl / Anzahl	DC resistance / Gleichstromwiderstand R_{DC} [Ω]	Temperature / Temperatur ϑ [°C]	Insulation resistance to iron / Isolationswiderstand gegen Eisen $R_{iso,\,60s}$ [MΩ]	PI	$U_{M,\,DC}$ [V]	Withstand voltage for 60 sec / Stehspannung für 60 s $U_{p,\,60s}$ [kV]
Stator system I / Phase / Ständer System I / Strang	4	0.00297	20	14900	n/a	500	2.4

ES 07 - Bearing run / Lagerlauf

Speed / Drehzahl 1200 rpm / 1/min at no-load / im Leerlauf

Temperature rise: Measuring point D-end / Temperaturerhöhung: Messstelle D-Seite ΔΘ [K]	17	Temperature rise: Measuring point ND-end / Temperaturerhöhung: Messstelle N-Seite ΔΘ [K]	49

ES 16 - Stator current symmetry / Ständer-Stromsymmetrie

Frequency / Frequenz f [Hz]	Voltage / Spannung U_1 [V]	Current Phase U / Strom Phase U $I_{1,U}$ [A]	Current Phase V / Strom Phase V $I_{1,V}$ [A]	Current Phase W / Strom Phase W $I_{1,W}$ [A]	electrical power / elektrische Leistung P_{el} [kW]	Power factor / Leistungsfaktor cos φ	DC voltage / Gleichspannung U_{DC} [V]	Remarks / Bemerkungen
40.0	218.2	730.6	716.9	724.9	9.3	0.034	n/a	converter supply at 88°C winding temperature, during heat run test

ES 11 / ES 21 - No-load test / Leerlaufprüfung

Frequency / Frequenz f [Hz]	Voltage / Spannung U_1 [V]	Current / Strom I_1 [A]	electrical power / elektrische Leistung P_{el} [kW]	Power factor / Leistungsfaktor cos φ	Exciter current / Erregerstrom I_f [A] / I_{e1} [A]	Speed / Drehzahl rpm / 1/min	Direction of rotation / Drehsinn	Remarks / Bemerkungen
40.0	513.9	17.8	3.8	0.23	n/a	1200.0	UVW	converter supply at 42°C winding temperature

ES 20 - Direction of rotation / Drehsinn

Connection / Anschluss L_1-L_2-L_3 to / an U-V-W → clockwise / rechts

ES 22 - Bearing housing vibrations / Lagergehäuseschwingungen

Speed / Drehzahl 1200.0 rpm / 1/min at no-load / im Leerlauf

	D-end / D-Seite			ND-end / N-Seite		
Coordinates / Richtung	x	y	z	x	y	z
Vibration displacement / Schwingweg s_{eff} [µm]	2.0	2.2	1.5	1.8	1.5	1.8
Vibration velocity / Schwinggeschwindigkeit v_{eff} [mm/s]	0.4	0.2	0.3	0.4	0.3	0.2
Vibration acceleration / Schwingbeschleunigung a_{eff} [m/s²]	0.3	0.3	0.2	0.3	0.3	0.2

EZ 03-06 - Check of temperature detectors / Prüfung von Temperatursensoren

Position / Ort	Type / Typ	Quantity / Anzahl	DC resistance / Gleichstromwiderstand R_{DC} [Ω]	Temperature / Temperatur ϑ [°C]	Insulation resistance to iron / Isolationswiderstand gegen Eisen $R_{iso,\,60s}$ [MΩ]	$U_{M,\,DC}$ [V]	Withstand voltage for 60 sec / Stehspannung für 60 s $U_{p,\,60s}$ [kV]
Winding / Wicklung	Pt100	6	100.0	0	30000	500	1.5
Bearing / Lager	Pt100	1/1	100.0	0	3000	100	n/a

☑ Witnessed ☐ Monitored ☐ Reviewed

Olaf Reger
25 Oct 2023
Hamburg Office
Lloyd's Register EMEA
LR426.2022

Issued: / Erstellt	Knöfel / Test bay engineer	Checked: / Geprüft	Gottschlich / Authorized inspection representative	Approved: / Genehmigt	LR O. Reger / Authorized inspection representative
ID number: / ID-Nummer	PB-K-2200120-45615803	Date: / Datum	25.10.2023	Revision: / Revision	0 from / vom 25.10.2023

Inspection certificate
Prüfbescheinigung

Inspection certificate 3.2 according to EN 10204
Abnahmeprüfzeugnis 3.2 nach EN 10204

Synchronous motor
Synchronmotor

EZ 16 - Check of space heater / Prüfung Stillstandsheizung

Quantity Anzahl	Rated data Bemessungsdaten U [V]	P [W]	DC resistance Gleichstromwiderstand R_{DC} [Ω]	Temperature Temperatur ϑ [°C]	Insulation resistance to iron Isolationswiderstand gegen Eisen $R_{iso, 60s}$ [MΩ]	$U_{M, DC}$ [V]	Withstand voltage for 60 sec Stehspannung für 60 s $U_{p, 60s}$ [kV]
1	*AC 230*	*150*	*353.4*	*20*	*30000*	*500*	*1.5*

EZ 18 - Check of speed encoder / Prüfung Drehzahlgeber

Quantity Anzahl	Manufacturer Hersteller	Type Typ	Serial no. Seriennummer	Pulses per rotation Pulse pro Umdrehung	Concentricity error Konusschlag [mm]
1	*Hübner*	*IGR FGHJ 2 HTL 2048*	*622546*	*2048*	*n/a*

Additional accessories / Weiteres Zubehör

1 x leckage monitoring cooling water
1 x shaft break DC 110 V, Typ 4BZFM 63

Remarks / Bemerkungen

Type test machine *LR HAM2303402/1 10 23*
Temperature rise test at nominal current, shaft voltage 1020 mV,
overspeed test 2640 rpm for 2 min

The machine was manufactured and tested in compliance with standard . The machine was found to be in order.
Die Maschine wurde hergestellt und geprüft in Übereinstimmung mit und für in Ordnung befunden.

Annex / Anlagen *n/a*

 45615803

☑ Witnessed ☐ Monitored ☐ Reviewed
Olaf Reger
25 Oct 2023
Hamburg Office
Lloyd's Register EMEA
LR426.2022

Issued: Erstellt	Knöfel Test bay engineer	Checked: Geprüft	Gottschlich Authorized inspection representative	Approved: Genehmigt	LR O. Reger Authorized inspection representative
ID number: ID-Nummer	PB-K-2200120-45615803	Date: Datum	25.10.2023	Revision: Revision	0 from vom 25.10.2023